

Determination of the Inverse Kinematics Branches of Solution Based on Joint Coordinates for Universal Robots-Like Serial Robot Architecture

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This paper introduces a classification of the inverse kinematics solutions (or robot postures) of six degrees-of-freedom serial robots with a geometry based on or similar to Universal Robots' arms. The solution of the inverse kinematics problem is first presented briefly, and the equations required to classify the robot postures (branches) based on the joint coordinates are then introduced. [DOI: 10.1115/1.4052805]

Keyword: theoretical kinematics

1 Introduction

Since the introduction of the first commercial collaborative robot (cobot) in 2008 by Universal Robots [1], the popularity of Cobots has increased exponentially and they now represent the fastest growing sector of industrial robotics. Most cobots, even from other manufacturers such as OMRON/TM and Hanwha, use the same geometry as the original UR5, the first commercialized cobot.

The DH parameters [2] of these Cobots include several zero and three twist angles (α_i) with value of $\pm\pi/2$, as presented in Table 1. The Distal (or Paul) variation of the DH parameters (see Ref. [3]) notation with a modification allowing negative a_i values is used in order to match the exact parameters given by Universal robots on both their website and their robots API. The reference frames used to define the DH parameters are shown in Fig. 1 and the relationship between the physical dimensions of the robot and the DH parameters is represented in Fig. 2.

For almost any robotic application, it is essential to find the solution of the inverse kinematic (IK) problem, i.e., to find the joint coordinates from the robot's flange pose (the term "pose" includes the position and the orientation). The methods that can be used to solve the IK are well known and documented. They can be divided into two categories, namely numerical methods (see Refs. [4–6] for example) and analytical methods (see Refs. [7–11]). The solution of the IK problem was also extended to other types of joints or redundant systems (see for instance [12–14]).

As presented in Sec. 2, this particular robot geometry yields an analytical solution with at most eight solutions. In order to ensure continuity in the joint trajectories, it is imperative to choose the correct solution because a change of solution branch would lead

to large discontinuities in the joint trajectories. A trivial solution to this problem when planning a full trajectory is to choose the solution with the coordinates that are the closest to the previous joint coordinates. This strategy is similar to what is done with a numerical approach, where an estimate is given to the algorithm and **only one solution is found, namely the solution that is the closest to the estimate.**

This strategy, however, cannot be used when performing the inverse kinematics on a single flange pose that is far from the current flange pose (no estimate available). The most common solution for this case is to generate a virtual Cartesian trajectory by interpolating linearly between the initial flange pose and the flange pose of interest and to perform the inverse kinematics at each point of this virtual trajectory, using the previous solution as an estimate for each point. **This method generally works well but offers no guarantee that the robot configuration will not change.** In practice, many configuration changes can occur when the initial and final flange poses polar coordinates are close to 180 deg from each other, as shown in Fig. 11. Also, this strategy is computationally ineffective. See Sec. 6.1 for an example of this situation.

Another challenge is to ensure that the correct solution is chosen when exiting a singularity. In this case, choosing the wrong solution will not lead to a discontinuous path but may put the robot in a configuration in which it cannot perform the requested task. For example, if the robot comes out of the elbow singularity with the elbow down, it is most probable that collisions will occur while doing any pick and place task on a surface corresponding to the base of the robot. See Sec. 6.2 for an example.

In summary, when programming robotic applications, it would be useful to have a method that allows the user/programmer to choose the robot configuration when solving the inverse kinematics and also to be able to determine the branches of solution directly from the joint coordinates. This way, it would be possible to know if the robot starts with the elbow up or down for example and to **force it to stay that way for the whole application process.** Should the robot start in the wrong branch of solution, it would be detected and the robot could change configuration by itself or inform the user that something is wrong.

Many papers showed how the different solutions of the IK equations can be associated with robot postures for a specific manipulator. For example, Ref. [15] solved the IK of the FANUC Arc Mate Sr. analytically, found an expression of the form $\theta_i = a \pm b$ and associated the + and – sign in the expression with posture of the arm, like elbow up or down, etc. This paper will do the same procedure for an UR-like robot but will also go further with the classification. Indeed, the procedure in Ref. [15] can classify the solution of the IK while solving it, but it cannot classify the posture of the robot directly from a given set of joint coordinates. The only way to obtain a similar result would be, for a given set of joint coordinates, to (1) solve the forward kinematic problem, (2) solve the IK for the resulting Cartesian position and classify the different solutions while doing so, (3) find the solution that corresponds to the initial set of joint coordinates, and (4) the classification of this solution is the one sought-after. This method is computationally ineffective compared to the one introduced in Sec. 5, which finds the classification directly from the joint coordinates. However, the

Table 1 DH parameters of UR-like robots

i	a_i	d_i	α_i
1	0	d_1	$\pi/2$
2	a_2	0	0
3	a_3	0	0
4	0	d_4	$\pi/2$
5	0	d_5	$-\pi/2$
6	0	d_6	0

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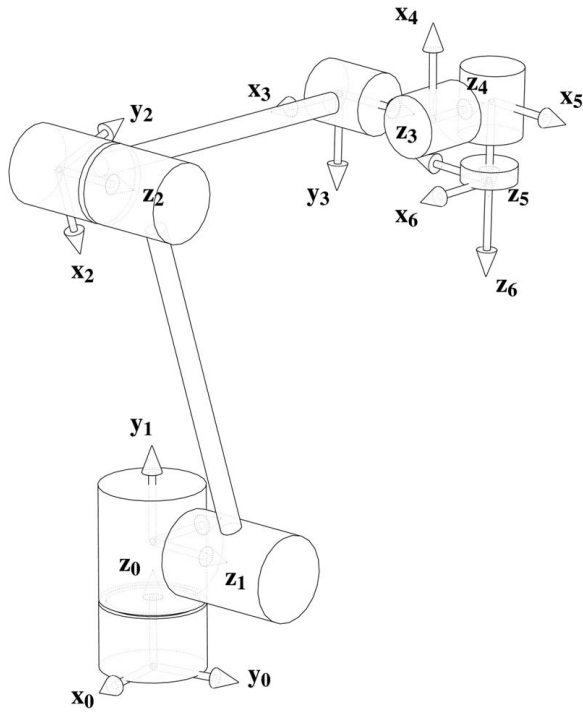


Fig. 1 DH frames

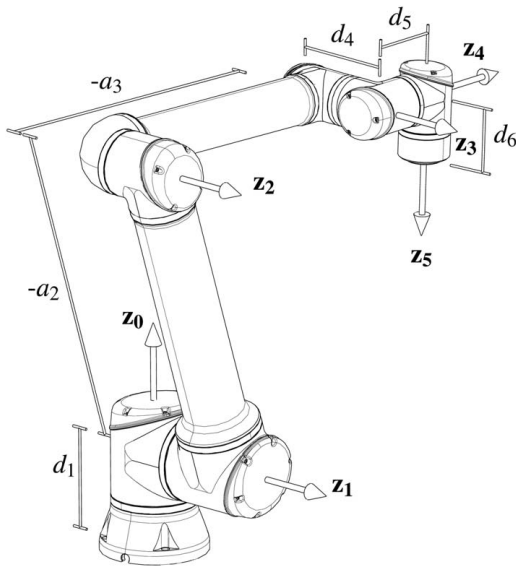


Fig. 2 DH parameters

four steps method described above was used to validate the results of Sec. 5.

2 Inverse Kinematics

The UR geometry is not strictly speaking wrist-partitioned (decoupled) [16–18] but it yields nonetheless a relatively simple analytical solution to the inverse kinematics problem. The eight solutions of the IK are already documented [7]. Without giving all the detailed equations, this section briefly recalls how the IK is solved since the classification of the branches of solution is obviously closely related to IK solving.

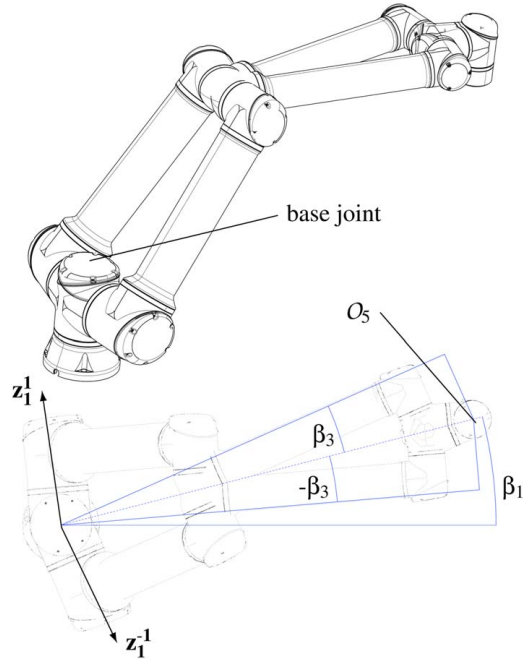


Fig. 3 Base joint (θ_1) solutions, isometric and top view of the robot

2.1 Solving for the Base Joint. The first step to solve the IK is to find the base joint coordinate. There are generally two solutions for this joint; an example with both solutions superimposed is shown in Fig. 3. A more complete geometric representation for the robot's configuration of Fig. 3 is shown in Fig. 4. Given the flange position ($\mathbf{p}_6$) and orientation ($\mathbf{Q}_6$), the position of $\mathcal{O}_5$ is found with

$$\mathbf{p}_5 = \mathbf{p}_6 - \mathbf{Q}_6[0 \ 0 \ d_6]^T \quad (1)$$

The base joint coordinate can then be expressed as

$$\theta_1 = \beta_1 + i \beta_2 + \pi/2, \quad i = -1, 1 \quad (2)$$

where

$$\beta_1 = \arctan 2(p_{5,y}, p_{5,x}) \quad (3)$$

and

$$\beta_2 = \frac{\pi}{2} - \beta_3 = \cos^{-1}\left(\frac{d_4}{r}\right) \quad (4)$$

Scalar value r is the distance between $\mathcal{O}_0$ and $\mathcal{O}_5$ in the $X_0 Y_0$ plane, i.e.,

$$r = \sqrt{p_{5,x}^2 + p_{5,y}^2} \quad (5)$$

Geometrically, angle β_1 can be described as the overall orientation of the robot or the direction of $\mathcal{O}_5$ relative to the base and angle β_2 is an offset associated with the fact that $\mathcal{O}_5$ does not lie in the $X_1 Y_1$ plane, i.e., d_4 is not zero.

When r is zero, there is a division by zero in Eqs. (3) and (4) and therefore, θ_1 is undefined. The robot is in the base-wrist singularity.

2.2 Solving for the Wrist 2–3 Joints. In order to find the second joint angle of the wrist (θ_5), the next step is to express the distance, projected on $\mathbf{z}_1$, between $\mathcal{O}_1$ and $\mathcal{O}_6$.

As shown in Fig. 5, this distance can be expressed as a function of θ_5 as

$$[\mathbf{p}_{6,z}]_1 = d_4 + d_6 \cos \theta_5 \quad (6)$$

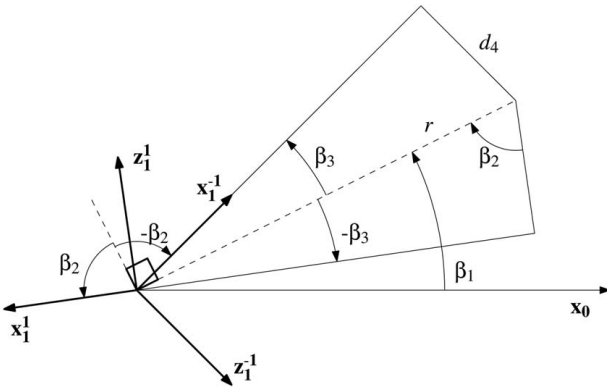


Fig. 4 Base joint (θ_1) solutions, top view geometry

and as a function of θ_1 and the Cartesian position ($\mathcal{O}_6$)

$$[\mathbf{p}_{6,z}]_1 = \mathbf{p}_{6,x} \sin \theta_1 - \mathbf{p}_{6,y} \cos \theta_1 \quad (7)$$

Combining Eqs. (6) and (7) leads to

$$\mathbf{p}_{6,x} \sin \theta_1 - \mathbf{p}_{6,y} \cos \theta_1 = d_4 + d_6 \cos \theta_5 \quad (8)$$

which can be solved for θ_5

$$\theta_5^{ij} = j \cos^{-1} \left(\frac{\mathbf{p}_{6,x} \sin \theta_1 - \mathbf{p}_{6,y} \cos \theta_1 - d_4}{d_6} \right) \quad (9)$$

yielding two solutions ($j = -1, 1$).

The four θ_1 – θ_5 combinations then lead to one wrist-3 angle value each (θ_6^j). See Ref. [7] for an explicit definition of the θ_6^j solutions.

2.3 Solving the Planar RRR Subchain. At this step, the positions of $\mathcal{O}_1$ and $\mathcal{O}_3$ are known. By exploiting the fact that the shoulder-elbow-wrist-1 joints form a planar RRR (three revolute joints) chain (see Fig. 6), the position of these joints can easily be found.

Using the cosine law, the distance $O_1 O_3$ (m) can be expressed a

$$m^2 = a_2^2 + a_3^2 - 2a_2a_3 \cos(\pi - \theta_3) \quad (10)$$

which can be solved for θ_3 as

$$\theta_3^k = k \cos^{-1} \left(\frac{m^2 - a_2^2 - a_3^2}{2a_2a_3} \right), \quad k = -1, 1 \quad (11)$$

Once θ_3 is known, finding θ_2 and θ_4 is trivial. The RRR planar chain is then solved, yielding two solutions (θ_{1-3}^{jk} , $k = -1, 1$), thus providing globally eight solutions for the robot IK, as shown in Table 2. A similar solution tree is presented in Ref. [8] for a Fanuc 200IC robot.

Indices i , j , and k are later referred to as inverse kinematics solution indices. Each of these indices can take two values, namely -1 or 1 and represents a branching (fork) in the inverse solution tree.

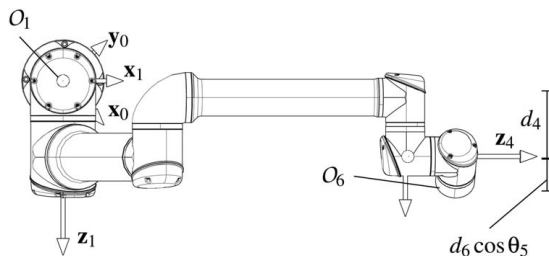


Fig. 5 Wrist2 (θ_5) geometry

Table 2 Inverse kinematics solutions tree structure. θ_1 : base joint, θ_5 : wrist1, θ_{1-3} : shoulder-elbow-wrist1 RRR planar chain

i	θ_1^{-1}		θ_1^1	
j	$\theta_5^{-1,-1}$	$\theta_5^{-1,1}$	$\theta_5^{1,-1}$	$\theta_5^{1,1}$
	$\theta_6^{-1,-1}$	$\theta_6^{-1,1}$	$\theta_6^{1,-1}$	$\theta_6^{1,1}$
k	$\theta_{1-3}^{-1,-1,-1}$	$\theta_{1-3}^{-1,-1,1}$	$\theta_{1-3}^{-1,1,-1}$	$\theta_{1-3}^{-1,1,1}$
	$\theta_{1-3}^{1,-1,-1}$	$\theta_{1-3}^{1,-1,1}$	$\theta_{1-3}^{1,1,-1}$	$\theta_{1-3}^{1,1,1}$

3 Qualitative Classification of the Inverse Kinematics Solutions

When describing a pose of the robot, the terms elbow up/down and wrist in/out are generally used. In addition to these two, the shoulder left/right must be added to completely describe a branch of solution of the IK.

3.1 Shoulder Left/Right. The shoulder side indicates, when looking from the robot base toward the wrist, on which side the shoulder joint is. The two configurations for a given flange pose can be seen in Fig. 6. The side of the shoulder generally has little impact on the application, unless there is an obstacle very close to the robot base.

3.2 Elbow Up/Down. Perhaps the most obvious classification criterion is the elbow of the robot, which can be either up or down, relative to the line passing from the shoulder to the wrist (Fig. 7). For most typical uses of Cobots, such as pick and place applications where the robot is fixed on a rigid surface, elbow up is preferred because otherwise the elbow would collide with the surface when trying to pick any part on that surface.

3.3 Wrist In/Out. This classification criterion refers to the position of the wrist relative to the flange position. If the wrist ($\mathcal{O}_4$) is closer to the base than the flange, the wrist is considered in and, otherwise, out (Fig. 8).

It is much more common to see robots in the wrist-in configuration since they have a greater reach this way but, for some trajectories where the flange needs to go close to the base, the wrist-out configuration can be useful. When the $\mathbf{z}_5$ axis is vertical, the in/out criterion is ill defined. In this situation, the wrist is rather up or down.

3.4 Wrist Right-/Left-Hand Rule. The wrist right-/left-hand rule criterion is similar to wrist in/out in the sense that it describes the position of the wrist. Wrist right/left refers to the right hand rule or left hand rule for the wrist joint axes. As shown in Fig. 9, right hand means that the wrist 2 joint axis can be found with the right hand rule, i.e.,

$$\mathbf{z}_4 = \mathbf{z}_3 \times \mathbf{z}_5 \quad (12)$$

where $\mathbf{z}_3$, $\mathbf{z}_4$, and $\mathbf{z}_5$ are the rotation axes of wrist1, wrist2, wrist3 joints, respectively, while left-hand rule means that the wrist2 joint axis can be found with the left-hand rule, i.e.,

$$\mathbf{z}_4 = -\mathbf{z}_3 \times \mathbf{z}_5 \quad (13)$$

The left/right wrist criterion is introduced because it is more closely related to the actual kinematic equations (as it will be shown in Sec. 5) than the wrist in/out criteria and therefore is easier to obtain directly from the joint coordinates.

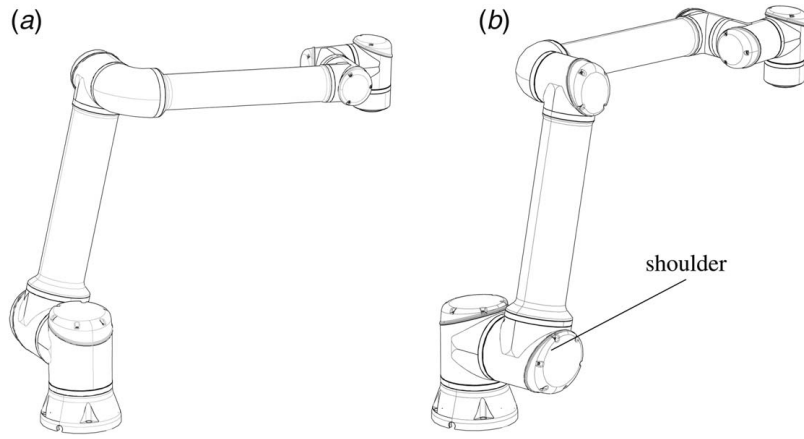


Fig. 6 Shoulder (a) left/(b) right solutions

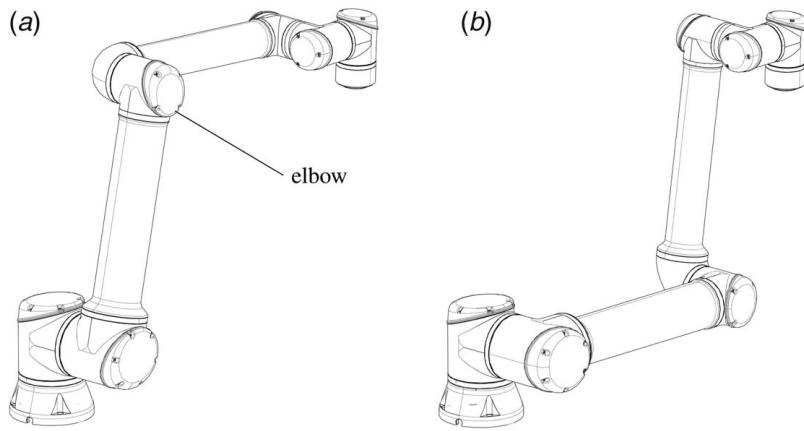


Fig. 7 Elbow (a) up/(b) down solutions

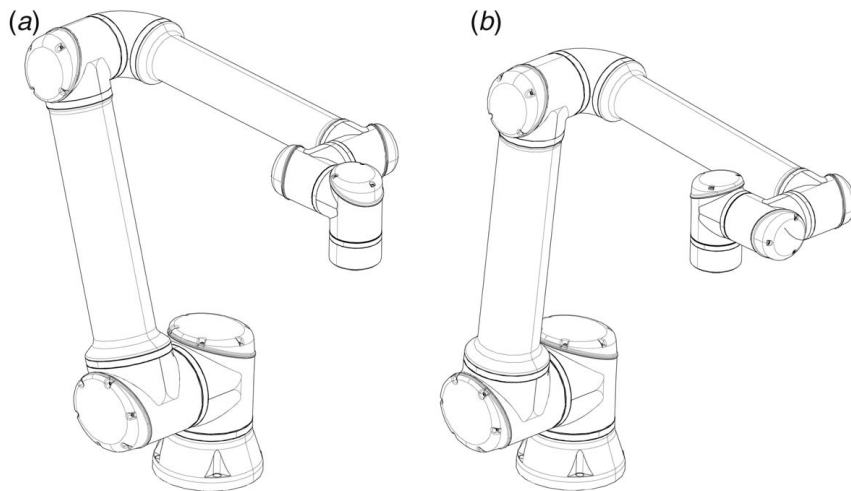


Fig. 8 Wrist (a) in/(b) out solutions

4 Relation With the Eight Inverse Kinematic Solutions

The relation between the solution classification criteria and the IK solution indices is shown in Table 3.

From the table, we can clearly see that index i , which is the base joint solution index, gives the shoulder side (left/right). Index j

gives the hand rule of the wrist (right-/left-hand rule) and finally index k is associated with the elbow side, but coupled with the shoulder side(i) such that the elbow side (up/down), g , is given by

$$g = i \cdot k \quad (14)$$

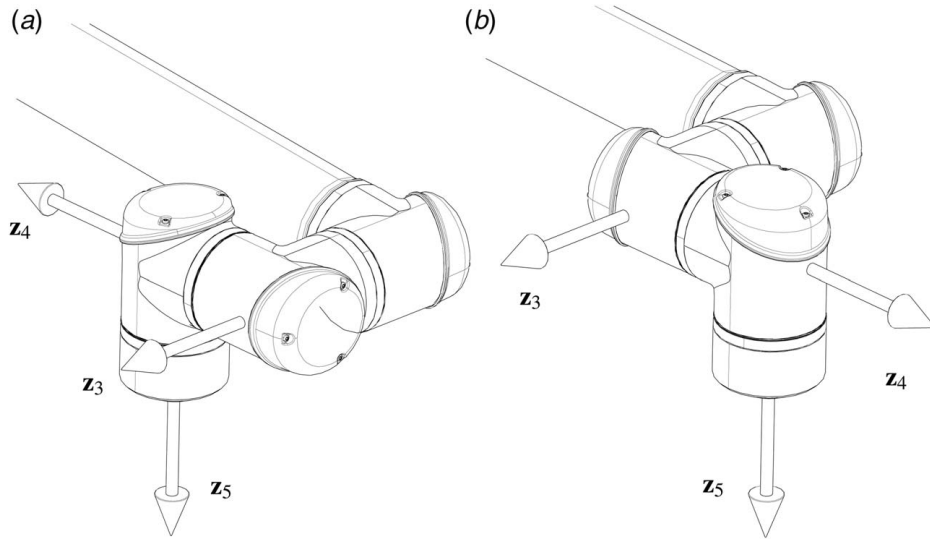


Fig. 9 Wrist (a) left-/(b) right-hand rule

with value -1 corresponding to elbow down. This relation can be explained by looking at Eq. (11), which shows that index k gives the sign of θ_3 and which corresponds to the elbow up/down configuration relative to the shoulder. In order to obtain the elbow configuration in the absolute reference frame, one must therefore multiply index k by the base joint solution index, i.e., i .

Finally, the wrist side (in/out), w , is given by the wrist hand rule (j) coupled with the shoulder side (i),

$$w = i \cdot j \quad (15)$$

with value -1 corresponding to wrist in.

5 Determination of the Inverse Kinematic Branches From the Joint Coordinates

In practice, the data available from the robot are the joint coordinates. It is therefore essential to be able to classify the IK branches from the joint coordinates in order for this classification to have any use in practice.

Two approaches can be used to find the solution indices from a given set of joint coordinates, namely

- (1) Solve the forward kinematic problem (obtain the flange pose), then find the n (up to 8) solutions of the IK problem and choose the solution corresponding to the robot's current joint coordinates. By keeping track of the solution indices used to find each solution, one can know which ones were used to find the solution corresponding to the current joint coordinates. This method is ineffective since it requires solving the IK even if the joint positions are already known.

Table 3 Truth table

n	i	j	k	Shoulder	Wrist	Elbow
1	1	1	1	Left	rh	Up
2	1	1	-1	Left	rh	Down
3	1	-1	1	Left	lh	Up
4	1	-1	-1	Left	lh	Down
5	-1	1	1	Right	rh	Down
6	-1	1	-1	Right	rh	Up
7	-1	-1	1	Right	lh	Down
8	-1	-1	-1	Right	lh	Up

- (2) Find the inverse method indices directly from the joint coordinates. This method requires the development of a set of equations in order to determine the IK indices. The rest of this section is devoted to obtaining these equations.

5.1 Base Side. By observation of Fig. 3, one can notice that the shoulder side (base joint solution index) can be found simply by considering the position of O_5 relative to the shoulder reference frame (O_1). This appears even more clearly on Fig. 10.

When the X_1 coordinate of O_5 is negative, the shoulder is on the left side. Algebraically, the base side index i can be expressed as

$$i = \text{signum} (a_2 \cos(\theta_3) + a_3 \cos(\theta_3 + \theta_4) + d_5 \sin(\theta_3 + \theta_4 + \theta_5)) \quad (16)$$

5.2 Wrist Hand Rule. The wrist hand rule index (j) can be found with

$$j = \text{signum} (z_5^T (z_4 \times z_6)) \quad (17)$$

which simplifies to

$$j = \text{signum} (\sin \theta_5) \quad (18)$$

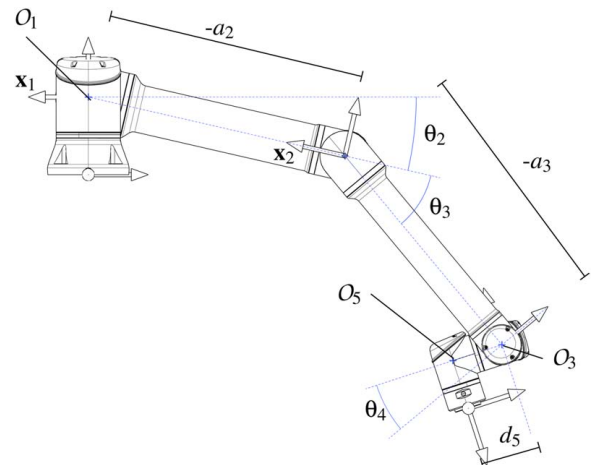


Fig. 10 Side view of the robot showing the planar RRR subchain

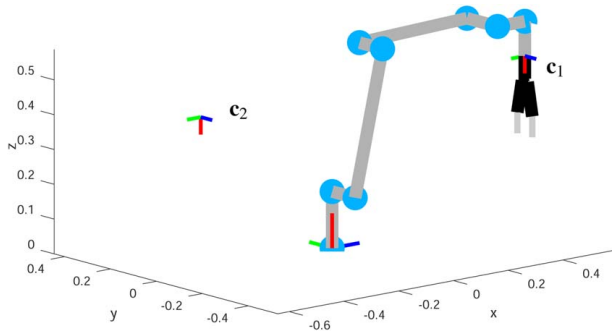


Fig. 11 Initial robot configuration and new single flange pose (c_2)

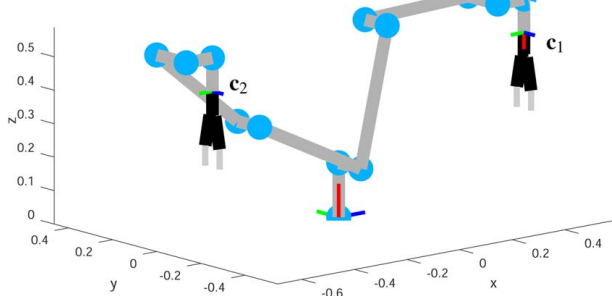


Fig. 12 Numerical solution

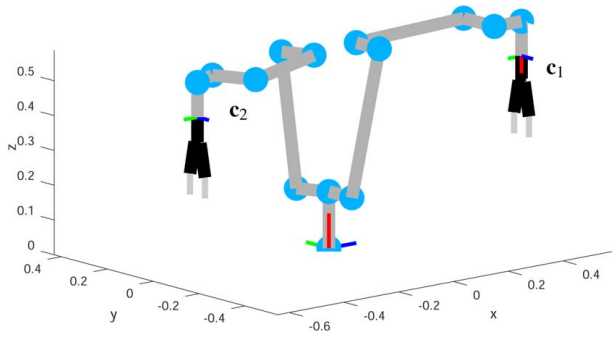


Fig. 13 Chosen analytic solution

From Eq. (9), one can notice that θ_5^1 solutions will always be in $[0; \pi] \pm 2n\pi$, $n=0, 1, 2, \dots$ range and θ_5^{-1} solutions in $[-\pi; 0] \pm 2n\pi$, $n=0, 1, 2, \dots$ range. These two ranges correspond precisely to $\text{signum}(\sin \theta)$ values of 1 and -1 , respectively, which confirms that Eq. (18) matches the values in Table 3.

5.3 Elbow Side. Finally, the elbow side index (k) can be found with

$$k = \text{signum}(\sin \theta_3) \quad (19)$$

This appears clearly on Fig. 10 and the same reasoning as for the wrist hand rule can be used to validate Eq. (19) since θ_3 is also found with an expression of the form $k \cdot \cos^{-1}(\dots)$.

6 Simulation Results

This section shows two real situations where the classification of the IK solutions can prevent problematic results.

6.1 Inverse Kinematic for an Isolated Flange Pose. In this example, the robot starts in a typical configuration. The starting joint coordinates are known, and the flange pose (c_1) can be evaluated with the forward kinematics. In this hypothetical situation, we want to know if the robot can pick a part at a location (c_2) that is on the other side of the robot base (as shown in Fig. 11). The position must be within the robot reach and there must be a collision-free robot configuration. It is therefore necessary to evaluate the IK for this single flange pose.

Using a numerical algorithm (described in Ref. [19]) to perform the IK, the solution shown in Fig. 12 is found in eight iterations. This solution is impractical for three reasons. First, the robot is almost in the elbow singularity, secondly, the elbow is now in the “down” configuration, which could cause collisions and finally, the wrist side has changed, which is hard to do in practice without collisions.

Using the analytical algorithm of Sec. 2, eight solutions are found. The solution that is the closest to the initial joint positions is exactly the same as the one found with the numerical algorithm (shown in Fig. 12), i.e., the impractical one. The closest solution is determined by subtracting the joint solution vectors to the initial joint configuration vector and wrapping the resulting joint angles to $[-\pi, \pi]$ before using the euclidean norm to evaluate the “distance” from the initial configuration.

On the other hand, the initial joint coordinates can be classified as shoulder-right, elbow-up and wrist-right using the equations of Sec. 5. This configuration can then be enforced during the IK using $i=-1$, $j=1$ and $k=-1$ such that the solution shown in Fig. 13 is found. This solution is more reasonable. It corresponds to the robot’s initial configuration with mostly a modification in the base joint position. In addition to the better final result, the computation time is lower because only one solution is found instead of 8 during the IK algorithm.

6.2 Exiting a Singularity. Assume now that the robot starts in the same configuration as in the previous situation. It performs a linear Cartesian trajectory toward the positive x -direction until it reaches the end of the robot’s workspace then reverses back to the starting position. When using either analytical and numerical IK algorithms, the side on which the elbow will exit the singular

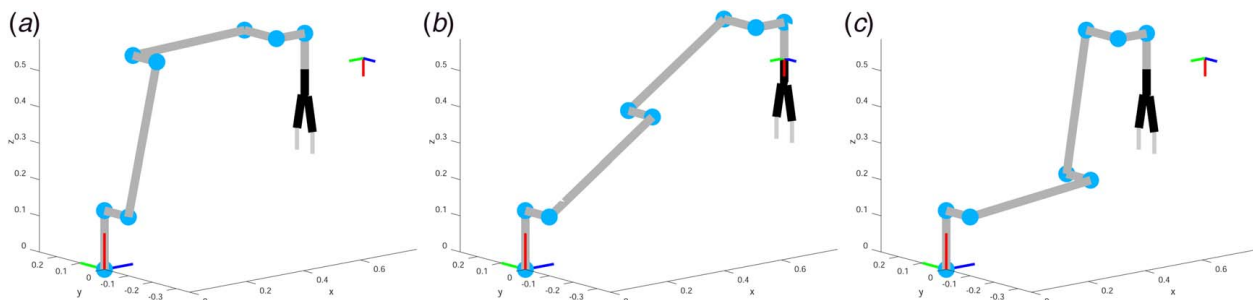


Fig. 14 Configuration change at singular position: (a) initial position, (b) singular position, and (c) post singular position

configuration is random due to numerical inaccuracies and joint control errors. Figure 14 shows an example of a configuration change when exiting a singular configuration. This situation can be avoided by using the proposed method. The initial configuration can be classified and enforced during the subsequent robot movements.

7 Conclusion

The eight solutions of the IK problem for a typical Cobot architecture can be classified using robot pose related criteria such as elbow up/down, shoulder left/right, or wrist in/out.

This paper proposed a method to classify the solution branches directly from a given set of joint positions (θ) with relatively simple mathematical expressions without having to solve the inverse kinematic problem.

When solving the IK, a given solution criterion can be specified in order to fully control which of the branches of solution is used and avoid discontinuous path or inadequate robot posture. Two examples were shown to demonstrate the effectiveness of the proposed method. The equations introduced in this paper can also be used to specify partially the robot posture classification while solving the inverse kinematics problem. For example, one could impose the elbow up posture but leave the other posture classification criterion free.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The authors attest that all data for this study are included in the paper. Data provided by a third party listed in Acknowledgment. No data, models, or codes were generated or used for this paper.

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